

Homework (Habanero)

Q1. What is the factored form of $x^3 + 8$?

- (A) $(x + 2)(x^2 - 2x + 4)$
- (B) $(x + 2)(x^2 + 2x + 4)$
- (C) $(x + 2)(x^2 - 2x - 4)$
- (D) $(x - 2)(x^2 + 2x + 4)$
- (E) $(x - 2)(x^2 - 2x - 4)$

Q2. A medicine dosage is modeled by $27x^3 - 1$. What is the factored form?

- (A) $(3x - 1)(9x^2 + 3x + 1)$
- (B) $(3x + 1)(9x^2 - 3x + 1)$
- (C) $(3x + 1)(9x^2 + 3x - 1)$
- (D) $(3x - 1)(9x^2 - 3x - 1)$
- (E) $(3x + 1)(9x^2 - 3x - 1)$

Q3. The growth of a cell is given by $x^3 - 64$. What is the factored form?

- (A) $(x - 4)(x^2 + 4x + 16)$
- (B) $(x + 4)(x^2 - 4x + 16)$
- (C) $(x + 4)(x^2 + 4x + 16)$
- (D) $(x - 4)(x^2 - 4x - 16)$
- (E) $(x + 4)(x^2 - 4x - 16)$

Q4. What is the factored form of $8x^3 + 27$?

- (A) $(2x + 3)(4x^2 - 6x + 9)$
- (B) $(2x - 3)(4x^2 + 6x + 9)$
- (C) $(2x + 3)(4x^2 + 6x - 9)$
- (D) $(2x - 3)(4x^2 - 6x - 9)$
- (E) $(2x + 3)(4x^2 - 6x - 9)$

Q5. The probability of a patient's recovery is modeled by

$x^3 - 125$. What is the factored form?

- (A) $(x - 5)(x^2 + 5x + 25)$
- (B) $(x + 5)(x^2 - 5x + 25)$
- (C) $(x + 5)(x^2 + 5x + 25)$
- (D) $(x - 5)(x^2 - 5x - 25)$
- (E) $(x + 5)(x^2 - 5x - 25)$

Q6. A biology experiment involves $64x^3 - 1$. What is the factored form?

- (A) $(4x - 1)(16x^2 + 4x + 1)$
- (B) $(4x + 1)(16x^2 - 4x + 1)$
- (C) $(4x + 1)(16x^2 + 4x - 1)$
- (D) $(4x - 1)(16x^2 - 4x - 1)$
- (E) $(4x + 1)(16x^2 - 4x - 1)$

Q7. The volume of a medical capsule is $x^3 + 216$. What is the factored form?

- (A) $(x + 6)(x^2 - 6x + 36)$
- (B) $(x - 6)(x^2 + 6x + 36)$
- (C) $(x - 6)(x^2 - 6x - 36)$
- (D) $(x + 6)(x^2 + 6x - 36)$
- (E) $(x + 6)(x^2 - 6x - 36)$

Q8. A statistical analysis involves $1 - 27x^3$. What is the factored form?

- (A) $(1 - 3x)(1 + 3x + 9x^2)$
- (B) $(1 + 3x)(1 - 3x + 9x^2)$
- (C) $(1 - 3x)(1 - 3x - 9x^2)$
- (D) $(1 + 3x)(1 + 3x - 9x^2)$
- (E) $(1 - 3x)(1 + 3x - 9x^2)$

Open Ended Questions (FRQ)

Q9. Factor $x^3 + 64$ and explain its relevance in a medical context.

Q10. The spread of a disease is modeled by $x^3 - 1000$. Factor this expression and describe a real-world scenario where this model is applicable.

Chef's Tips

- Read each question carefully and choose the correct answer.
- For free response questions, support your answers with relevant calculations and explanations.
- Use the space provided for the free response questions to show your work and write your answers clearly.